厦门大学《微积分 I-2》课程期中试卷

试卷类型：（理工类 A 卷） 考试日期

一、 i). $\overrightarrow{AB}\times\overrightarrow{AC}=(-18,24,18)$ ii) $V_{平行六面体}=\left(\overrightarrow{AB},\overrightarrow{AC},\overrightarrow{AD}\right)=\left(\overrightarrow{AB}\times\overrightarrow{AC}\right)\cdot\overrightarrow{AD}=6$

二、 $1.\tan(x+y)=x+C$ （思路：令$x+y=u$,则原方程化为:$\sec^2 u du=dx$）

$2.y=\dfrac{1}{1-x}$ （思路：i)令$y'=P(y)$,则$y''=P\dfrac{dP}{dy}$,

$\therefore$原方程化为：$PdP=2y^3dy$

解得：$P^2=y^4+C_1$,

ii)把初始条件代入得：$C_1=0.$

$\therefore P=y^2.$

iii)解得:$\dfrac{dy}{dx}=y^2$得:$-\dfrac{1}{y}=x+C_2$

三、 $f(x)=(1+x)e^x$

（思路：i)两边对x求导得:$f'(x)-f(x)=e^x$且初始条件$f(0)=1$

ii)令$f(x)=y$,则$y'-y=e^x\cdots(1)$

1)对应齐次方程为：$y'-y=0$解得：$y=Ce^x$

2)(常数变易法)可设(1)的通解为$y=u(x)e^x.$

代入解得$u(x)=x+C_1.$

四、 $y=(C_1+C_2x)e^x+\dfrac{1}{2}\cos x$

（思路：i)对应齐次方程$y''-2y'+y=0\cdots(2)$ 特征值：$\lambda_1=\lambda_2=1.$

两线性无关特解$y_1=e^x,y_2=xe^x$ $\therefore$(2)的通解为$Y=C_1e^x+C_2xe^x$

ii) 1)$y''-2y'+y=1\cdots(1')$ $\because f_1(x)=1,(\lambda=0)$

$\lambda=0$不是特征值，可设(1′)的一个特解为$y_1^*=A_0$ 解得：$A_0=1$

2)$y''-2y'+y=\sin x\cdots(1''')$ $\because f_1(x)=1,(\lambda=0,\omega=1)$

$\lambda+i\omega=i$不是特征值，可设(1′)的一个特解为$y_1^*=B_0\sin x+D_0\cos x$

解得：$B_0=0,D_0=\dfrac{1}{2}$

五、 $d=\dfrac{1}{\sqrt{3}}$

(思路：i)过$x-1=y=\dfrac{z}{2}$且平行于$\dfrac{x}{1}=\dfrac{y}{2}=\dfrac{z}{3}$的平面$\pi$：$x-y+z-1=0$

关键：可设过$L_1: x-1=y=\dfrac{z}{2}\Leftrightarrow\begin{cases}x-y-1=0\\2y-z=0\end{cases}$的

平面π：$\lambda(x-y-1)+\mu(2y-z)=0$

$ii)$ $\dfrac{x}{1}=\dfrac{y}{2}=\dfrac{z}{3}$上点$O(0,0,0)$到平面$\pi$的距离$d=\dfrac{|0-0+0-1|}{\sqrt{1^2+1^2+1^2}}=\dfrac{\sqrt{3}}{3}$

六、：$\dfrac{\partial(\rho,\theta)}{\partial(x,y)}=\dfrac{\partial\rho}{\partial x}\dfrac{\partial\theta}{\partial y}-\dfrac{\partial\rho}{\partial y}\dfrac{\partial\theta}{\partial x}=\dfrac{1}{ab\rho}$

(思路一：$i)\begin{cases}x=a\rho\cos\theta\\y=b\rho\sin\theta\end{cases}\Rightarrow\begin{cases}\rho=\dfrac{\sqrt{b^2x^2+a^2y^2}}{ab}\\\theta=\arctan\dfrac{ay}{bx}\end{cases}$

$$\Rightarrow\begin{cases}\dfrac{\partial\rho}{\partial x}=\dfrac{bx}{a\sqrt{b^2x^2+a^2y^2}}=\dfrac{\cos\theta}{a},\\\dfrac{\partial\theta}{\partial x}=\dfrac{1}{1+\left(\dfrac{ay}{bx}\right)^2}\cdot\left(-\dfrac{ay}{bx^2}\right)=-\dfrac{aby}{\left(b^2x^2+a^2y^2\right)}=-\dfrac{\sin\theta}{a\rho}\\\dfrac{\partial\rho}{\partial x}=\dfrac{ay}{b\sqrt{b^2x^2+a^2y^2}}=\dfrac{\sin\theta}{b},\\\dfrac{\partial\theta}{\partial y}=\dfrac{1}{1+\left(\dfrac{ay}{bx}\right)^2}\cdot\left(\dfrac{a}{bx}\right)=\dfrac{abx}{\left(b^2x^2+a^2y^2\right)}=\dfrac{\cos\theta}{b\rho}\end{cases}$$

$ii)$ 求$Jacobi$行列式$\dfrac{\partial(\rho,\theta)}{\partial(x,y)}=\begin{vmatrix}\dfrac{\partial\rho}{\partial x}&\dfrac{\partial\theta}{\partial x}\\\dfrac{\partial\rho}{\partial y}&\dfrac{\partial\theta}{\partial y}\end{vmatrix}=\dfrac{\partial\rho}{\partial x}\cdot\dfrac{\partial\theta}{\partial y}-\dfrac{\partial\theta}{\partial x}\cdot\dfrac{\partial\rho}{\partial y}=\dfrac{1}{ab\rho}$

(思路二：$i)\begin{cases}x=a\rho\cos\theta\\ y=b\rho\sin\theta\end{cases}\Rightarrow\begin{cases}\dfrac{\partial x}{\partial\rho}=a\cos\theta,\dfrac{\partial x}{\partial\theta}=-a\rho\sin\theta\\ \dfrac{\partial y}{\partial\rho}=b\sin\theta,\dfrac{\partial y}{\partial\theta}=b\rho\cos\theta\end{cases}$

$ii)$ 求*Jacobi*行列式$\dfrac{\partial(x,y)}{\partial(\rho,\theta)}=\begin{vmatrix}\dfrac{\partial x}{\partial\rho}&\dfrac{\partial x}{\partial\theta}\\ \dfrac{\partial y}{\partial\rho}&\dfrac{\partial y}{\partial\theta}\end{vmatrix}=\dfrac{\partial x}{\partial\rho}\cdot\dfrac{\partial y}{\partial\theta}-\dfrac{\partial x}{\partial\theta}\cdot\dfrac{\partial y}{\partial\rho}=ab\rho$

$iii)$*Jacobi*行列式$\dfrac{\partial(\rho,\theta)}{\partial(x,y)}=\dfrac{1}{\dfrac{\partial(x,y)}{\partial(\rho,\theta)}}=\dfrac{1}{ab\rho}$

七、 $\dfrac{\partial z}{\partial x}=f_1'+\dfrac{1}{y}f_2',\quad \dfrac{\partial^2 z}{\partial x\partial y}=-f_{11}''-\dfrac{x+y}{y^2}f_{12}''-\dfrac{1}{y^2}f_2'-\dfrac{x}{y^3}f_{22}''$

(思路：$i)$令$\begin{cases}x-y=u\\ \dfrac{x}{y}=v\end{cases}$

$ii)$ $\dfrac{\partial z}{\partial x}=\dfrac{\partial z}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\dfrac{\partial z}{\partial v}\cdot\dfrac{\partial v}{\partial x}$

$iii)\dfrac{\partial^2 z}{\partial x\partial y}=\dfrac{\partial}{\partial y}\left(\dfrac{\partial z}{\partial x}\right)=\dfrac{\partial}{\partial y}\left(f_1'+\dfrac{1}{y}f_2'\right)=\dfrac{\partial f_1'}{\partial y}+\left(-\dfrac{1}{y^2}f_2'+\dfrac{1}{y}\dfrac{\partial f_2'}{\partial y}\right)$

而$\dfrac{\partial f_1'}{\partial y}=\dfrac{\partial f_1'}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\dfrac{\partial f_1'}{\partial v}\cdot\dfrac{\partial v}{\partial x}=-f_{11}''-\dfrac{1}{y^2}f_{12}''$

$\dfrac{\partial f_2'}{\partial y}=\dfrac{\partial f_2'}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\dfrac{\partial f_2'}{\partial v}\cdot\dfrac{\partial v}{\partial x}=-f_{21}''-\dfrac{1}{y^2}f_{22}''$

八、 1. 投影曲线方程：$\begin{cases}\dfrac{x^2}{5}+\dfrac{y^2}{4}=1\\ z=0\end{cases}$

(思路：$i)$交线$\Gamma:\begin{cases}x^2+y^2+\dfrac{z^2}{4}=5\\ y=z\end{cases}\Leftrightarrow\begin{cases}\dfrac{x^2}{5}+\dfrac{y^2}{4}=1\\ y=z\end{cases}$

2. $2\sqrt{2},\quad\sqrt{5}$

(思路：*ii*) 1)$\forall M(x,y,z)\in$交线Γ,则$d^2=|OM|=x^2+y^2+z^2$

本题问题转化为求解$f(x,y,z)=x^2+y^2+z^2$在限制条件

$$\frac{x^2}{5}+\frac{y^2}{4}=1\text{与}y=z\text{下的最值}$$

2)令$F(x,y,z,\lambda,\mu)=x^2+y^2+z^2+\lambda\left(4x^2+5y^2-20\right)+\mu\left(y-z\right)$

九、 切线方程为$\frac{x}{2}=\frac{y-2}{1}=\frac{z-1}{1}$　　法平面方程：$2x+y+z-3=0$

(思路：关键：$\Gamma:\begin{cases}x=x\\ y=\psi(x)\\ z=\omega(x)\end{cases}$ (x为参数)，则切线L的方向$\vec{v}=(1,\psi'(x),\omega'(x))\big|_{x=x_0}$

i) $\because 3x^2+y-z-1=0\Rightarrow 6xdx+dy-dz=0\cdots(1)$

ii)$\because x-y^2+2z+2=0\Rightarrow dx-2ydy+2dz=0\cdots(2)$

(1)(2)解得：$dy=\frac{12x+1}{2y-2}dx,\quad dz=\frac{12xy+1}{2y-2}dx$

(思路二：*i*) 曲线$\Gamma\subset$曲面$S_1:F(x,y,z)=0$,其中$F(x,y,z)=3x^2+y-z-1$

$\therefore F'_x=6x,\quad F'_y=1,\ F'_z=-1$

$\therefore$曲面S_1在点$P_0(0,2,1)$处的切平面的

法向量$\vec{n}_1=(F'_x,F'_y,F'_z)\big|_{P_0}=(6x,1,-1)\big|_{x=0}$

ii)曲线$\Gamma\subset$曲面$S_2:G(x,y,z)=0$,其中$G(x,y,z)=x-y^2+2z+2=0$

$\therefore G'_x=1,\quad G'_y=-2y,\ G'_z=2$

$\therefore$曲面S_2在点$P_0(0,2,1)$处的切平面的

法向量$\vec{n}_2=(G'_x,G'_y,G'_z)\big|_{P_0}=(1,-2y,2)\big|_{y=2}$

iii) $\therefore$曲线Γ过P_0的切线L 的方向$\vec{v}=\vec{n}_1\times\vec{n}_2=(-2,-1,-1)$

十、 **(1)**不可微

(思路：*i*) 求$\frac{\partial f}{\partial x}(0,0)=0,\frac{\partial f}{\partial y}(0,0)=0$

ii)$\Delta z=f(0+\Delta x,0+\Delta y)-f(0,0)=\frac{\Delta x\cdot(\Delta y)^2}{(\Delta x)^2+(\Delta y)^2}$

iii)求 $\lim\limits_{(\Delta x,\Delta y)\to(0,0)} \dfrac{\Delta z - f_x'(0,0)\Delta x - f_y'(0,0)\Delta y}{\rho} = \dfrac{1}{2\sqrt{2}} \neq 0$

$\therefore \Delta z - f_x'(0,0)\Delta x - f_y'(0,0)\Delta y \neq o(\rho)$

$\therefore f(x,y)$在$(0,0)$不可微

(2)

i) 以$(0,0)$为始点, 沿方向$(\cos\alpha,\sin\alpha)$的射线l: $\begin{cases} x = \rho\cos\alpha \\ y = \rho\sin\alpha \end{cases}$

ii) $f(x,y)$在点$(0,0)$沿方向l的方向导数

$$\frac{\partial f}{\partial l}\Big|_{(0,0)} = \lim_{\rho\to 0} \frac{f(0+\Delta x, 0+\Delta y) - f(0,0)}{\rho}$$

其中$\Delta x = \rho\cos\alpha, \Delta y = \rho\cos\alpha$

iii) $f(0+\Delta x,0+\Delta y) - f(0,0) = \dfrac{\Delta x\cdot(\Delta y)^2}{(\Delta x)^2+(\Delta y)^2} = \dfrac{\rho^3\cos\alpha\cdot\sin^2\alpha}{\rho^2}$

iv) $\therefore \dfrac{\partial f}{\partial l}\Big|_{(0,0)} = \lim\limits_{\rho\to 0}\cos\alpha\cdot\sin^2\alpha = \cos\alpha\cdot\sin^2\alpha$

$\therefore f(x,y)$在点$(0,0)$沿方向l的方向导数方向导数是存在的,

且$\dfrac{\partial f}{\partial l}\Big|_{(0,0)} = \cos\alpha\cdot\sin^2\alpha$

十一、

(思路: *i)* 在极坐标下, $f(x,y) = f(x(\rho,\theta), y(\rho,\theta))$, 其中: $\begin{cases} x = \rho\cos\theta \\ y = \rho\sin\theta \end{cases}$

ii) $\dfrac{\partial f}{\partial \rho} = \dfrac{\partial f}{\partial x}\cdot\dfrac{\partial x}{\partial \rho} + \dfrac{\partial f}{\partial y}\cdot\dfrac{\partial y}{\partial \rho} = \dfrac{\partial f}{\partial x}\cdot\cos\theta + \dfrac{\partial f}{\partial y}\cdot\sin\rho$

$\therefore \rho\dfrac{\partial f}{\partial \rho} = x\dfrac{\partial f}{\partial x} + y\dfrac{\partial f}{\partial y}\cdots\cdots(1)$

iii).由(1)与已知得: $\lim\limits_{\rho\to+\infty}\rho\dfrac{\partial f}{\partial \rho} = 1$

$\therefore$由极限的保号性, 存在R_0,当$\rho > R_0$时, 有

$\rho\dfrac{\partial f}{\partial \rho} \geq \dfrac{1}{2}$,即: $\dfrac{\partial f}{\partial \rho} \geq \dfrac{1}{2\rho}\cdots\cdots(2)$

iv) $\because \dfrac{\partial f(\rho,\theta)}{\partial \rho} \geq \dfrac{1}{2\rho}\cdots\cdots(2)$ $\therefore \int_{R_0}^{R}\dfrac{\partial f(\rho,\theta)}{\partial \rho}d\rho \geq \int_{R_0}^{R}\dfrac{1}{2\rho}d\rho$ $(R \geq R_0)$

$\therefore f(\rho,\theta)\big|_{R_0}^{R} \geq \frac{1}{2}(\ln R - \ln R_0)$

$\therefore f(R,\theta) - f(R_0,\theta) \geq \frac{1}{2}(\ln R - \ln R_0)$ $\quad \therefore \lim_{R\to+\infty} f(R,\theta) = +\infty$

即：$\lim_{\rho\to+\infty} f(\rho,\theta) = +\infty$

v)∵ f在全平面上有连续的一阶偏导数,∴ f在全平面上连续

∴ f在闭区域$x^2 + y^2 \leq 1$上连续

∴ f在闭区域$x^2 + y^2 \leq 1$上最小值m

vi)∵ $\lim_{\rho\to+\infty} f(\rho,\theta) = +\infty$,存在$\rho_0$,当$\rho > \rho_0$时，$f(\rho,\theta) \geq m$

这就说明了f只能在$x^2 + y^2 \leq \rho_0^2$上取得其最小值

vii)∵ f在闭区域$x^2 + y^2 \leq \rho_0^2$上连续,

∴ f在闭区域$x^2 + y^2 \leq \rho_0^2$上可取得其最小值